

CS 432/532, Fall 2026 Syllabus

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Course Information

- **Instructor:** Nasreen Arif
- **Course title, number, section:** Web Science CS432/532
- **Semester and year:** Fall 2026
- **Delivery mode:** Asynchronous online

Instructor Contact and Office Hours

Email: nasreen at cs.odu.edu **Website:** <https://www.cs.odu.edu/~narif>

Office hours will be held on Mondays and Wednesdays by appointment only. Please use the appointment-scheduling link provided in Canvas. All scheduled meetings will take place via Zoom. Once your appointment is confirmed, you will receive a Zoom link in the confirmation email.

Course Description

Catalog Description

Provides an overview of the World Wide Web and associated decentralized information structures, focusing mainly on the computing aspects of the Web: how it works, how it is used, and how it can be analyzed. Students will examine a number of topics including web architecture, web characterization and analysis, web archiving, Web 2.0, social networks, collective intelligence, search engines, web mining, information diffusion on the Web, and the Semantic Web.

Course Description

The Web has fundamentally changed how we learn, play, communicate, and work. Its influence has become so monumental that it has given birth to a new science: Web Science, or the science of decentralized information structures. Although Web Science is interdisciplinary by nature, this course will focus mainly on the computing aspects of the Web.

Prerequisites

A grade of C or better in any one of: CS 250, CS 251, CS 253, CS 260, CS 261, CS 263, or DASC 255

Programming Background

We will use Python throughout the course. You are not required to know Python before the course begins, but because previous programming experience is required, you are expected to learn the necessary syntax quickly. If you are unfamiliar with Python, you should review introductory Python material before beginning the programming assignments.

Recommended introductory resources include:

- [CS 1110: Introduction to Computing Using Python](#) (Cornell University)
- [A Gentle Introduction to Programming Using Python](#) (MIT)
- [Python in One Easy Lesson](#) by Nick Parlante

Required Materials

Required Textbook

- There is no required textbook

Supplemental Materials (Optional)

Here is a list of recommended books:

- [Python in a Nutshell](#) - 3rd edition available [via ODU](#)
 - mentioned in Module 2
- [Learning Python](#) - 5th edition available [via ODU](#)
 - mentioned in Module 2
- [Search Engines Information Retrieval in Practice](#) - pdf available
 - used in Module 3
- [Introduction to Information Retrieval](#) - pdf available
 - used in Module 3
- [Networks, Crowds, and Markets: Reasoning About a Highly Connected World](#) - preprint available
 - used in Module 5
- [Programming Collective Intelligence](#) - available [via ODU](#)
 - used in Module 6
- [Speech and Language Processing](#) - pdf available
 - mentioned in Module 8

Technology Requirements

- Hardware:
 - Computer or laptop (Windows, macOS, or Linux) capable of running modern web browsers and development tools
 - Reliable high-speed Internet connection
 - Microphone and speakers/headset for accessing recorded lectures and participating in online activities
 - Webcam (optional, recommended for virtual office hours if needed)

- Software:
 - Canvas LMS (course management and submissions)
 - Modern web browser (latest version of Chrome, Firefox, or Edge)
 - Visual Studio Code (VS Code)
 - Microsoft 365 (or equivalent word-processing)
 - GitHub account (for version control, collaboration, and assignment submission)
 - Access to ODU CS Linux servers (CS account required)
 - Google Colab (for cloud-based execution of programming notebooks)

Course Learning Objectives

After completing this course, you should be able to do the following:

- CLO1: Explain the interdisciplinary nature of Web Science.
- CLO2: Explain the main components of web architecture.
- CLO3: Explain how search engines index and rank webpages given a query.
- CLO4: Write Python code to interact with various web APIs and services, including web servers, web archives, and Memento aggregators.
- CLO5: Explain the impact of social networks on how information and disinformation spread on the Web.
- CLO6: Explain the differences between classification and clustering algorithms.
- CLO7: Identify appropriate machine-learning algorithms to apply to a given problem.

Grading Criteria

In this course, I will calculate your final grade based on the following categories:

Assignments	Percentage/Points
Learning Check	5%
Online Discussion	5%
Homework	90%

Assignment Requirements

Getting Started Activities – Required / Extra Credit

The Getting Started module includes required activities that students must complete before beginning the regular course modules. These activities may include reviewing the syllabus, confirming access to Canvas and required course tools, setting up GitHub or other required accounts, and reviewing assignment submission expectations.

The Getting Started module may also include optional extra credit activities. These activities are not required, but students who choose to complete them must submit them by the separate due dates listed in Canvas. Extra credit activities may have different deadlines from the regular weekly module assignments.

Assignment Category 1: Learning Check – 5%

Purpose: Learning Checks ensure students understand foundational terminology and conceptual material before progressing to applied assignments

Course Learning Objectives: CLO1, CLO2, CLO3, CLO5, and CLO6.

Frequency: Short auto-graded quizzes in Modules 1–8

Instructions:

- Complete the short Canvas quiz.
- Questions may include multiple-choice and short-answer items.
- The number of attempts may be limited as specified in Canvas.

Deliverables: Submit directly through Canvas Quiz tool.

Evaluation Criteria: Completion by deadline

Assignment Category 2: Online Discussions – 5%

Purpose: Discussions develop critical thinking about social, ethical, and interdisciplinary dimensions of the Web.

Course Learning Objectives: CLO1, CLO2, CLO3, CLO5, and CLO6.

Frequency: Modules 1–8

Instructions:

Initial Post Requirements:

- Write response in complete sentences using proper grammar and academic tone.
- Address all parts of the discussion prompt.
- Demonstrate understanding of relevant course concepts and terminology.
- Apply technical, social, ethical, or interdisciplinary perspectives where appropriate.
- Provide a clear concluding statement that synthesizes your main point.
- Submit the initial post by **Friday at 11:59 PM**

Peer Response Requirements:

- Reply directly to a classmate's original post.
- Write in complete, professional sentences.
- Extend the discussion by:
 - Adding new analysis,
 - Introducing an additional concept,
 - Offering a constructive counterpoint, or
 - Posing a thoughtful analytical question.
- Responses such as "Good job" or "I agree" without substantive contribution will not receive credit.
- Maintain respectful and professional communication, even when disagreeing.
- Submit peer responses by Sunday at 11:59 PM.
- Students are expected to provide follow-up responses in more than half of the weekly discussions

Deliverables: Canvas Discussion Board submission

Evaluation Criteria:

- Depth of analysis
- Integration of readings
- Professional engagement
- Clarity and structure

Assignment Category 3: Homework – 90%

Purpose: Homework assignments serve as the primary assessment component of this course. These assignments require students to apply theoretical concepts from Web Science to practical computational problems. Students will analyze web structures, retrieve and process web data, implement algorithms, and evaluate information systems using Python and related tools. The goal is to demonstrate technical competency, analytical reasoning, and the ability to connect conceptual understanding with implementation.

Course Learning Objectives: CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, and CLO7.

Frequency: Five major homework assignments distributed across the semester. Each assignment builds progressively on prior modules

Instructions:

For each homework assignment, students are expected to:

- Carefully review the assignment specification.
- Implement required programs using Python.
- Apply relevant web science concepts (e.g., crawling, indexing, ranking, network analysis, clustering).
- Analyze results and interpret findings.
- Document methodology, assumptions, and conclusions.
- Properly cite all references used, including AI tools if applicable.
- Commit all code to the designated GitHub repository before the deadline

Deliverables: GitHub link on Canvas

Evaluation Criteria: Homework assignments are evaluated using the following criteria:

- Technical Correctness
- Conceptual Understanding
- Code Quality and Organization
- Analysis and Discussion
- Documentation and Professionalism

There is no midterm or final examination.

Grading Scale

The grading scale for undergraduate students (CS 432) is as follows:

Percentage	Letter Grade
100 – 92	A
91 – 90	A-
89 – 87	B+
86 – 82	B
81 – 80	B-
79 – 77	C+
77 – 72	C
71 – 70	C-
69 – 67	D+
66 – 62	D
61 – 60	D-
59 – 00	F

The grading scale for graduate students (CS 532) is as follows:

Percentage	Letter Grade
100 – 94	A
93 – 90	A-
89 – 88	B+
87 – 84	B
83 – 80	B-
79 – 78	C+
77 – 74	C
73 – 70	C-
69 – 00	F

Policy on Incomplete Grades

A grade of "I" indicates assigned work yet to be completed in a given course, or absence from the final examination, and is assigned only upon instructor approval of a student request. The "I" grade may be awarded only in exceptional circumstances beyond the student's control. The "I" grade becomes an "F" if not removed by the day grades are due for the following term, based on specific criteria: [Incomplete, Withdraws and Z grades](#).

Expectations

What you can expect from me, as your instructor:

- I will ensure the accuracy of the course content.
- I will provide clear guidelines regarding the course assignments.
- I will be available to answer your questions.
- I will provide meaningful feedback in a timely manner.
- I will do my best to create a worthwhile learning experience for all my students.
- I will follow fair and clear evaluation guidelines for all the course assignments.
- I will do my best to prepare you for assignments.

What I expect from you, as my student:

- You will complete all course readings and assignments in a timely manner.
- You will follow the ODU honor pledge.
- You will follow proper [netiquette](#).
- You will interact with your faculty and classmates professionally and respectfully.
- You will avoid sarcasm and inappropriate language, including the use of ALL CAPS.
- You will acknowledge your classmates' ideas and build on them to contribute to the discussion.

Related Courses

- All things HTTP — [CS 531: Web Server Design](#)
- Information retrieval and metadata — [CS 734/834: Introduction to Information Retrieval](#)
- Visualization and analytics — [CS 625: Data Visualization](#) and [CS 725/825: Information Visualization](#)
- Web programming — [CS 418/518: Web Programming](#)
- Data science — [CS 620: Introduction to Data Science](#)

Course History

This course at ODU has been taught by [Dr. Michele Weigle](#) and was originally developed by [Dr. Michael Nelson](#), based on ODU alumnus Dr. Frank McCown's Introduction to Web Science course at Harding University. In previous semesters, it has also been taught by Dr. Alexander Nwala, a former ODU doctoral student.

Course Policies

Email and Canvas

Each student must check ODU email and the course Canvas site daily. Use the course Canvas Discussion Board to ask and answer general course-related questions. Important updates, including assignment deadline changes and office-hour cancellations, will be posted through Canvas Announcements.

Attendance and Participation

This course is delivered in an asynchronous format. There are no required live class meetings; however, consistent and active engagement is required for academic success

Make-up Work

Make-up opportunities for graded activities are available only with a valid written medical or university excuse. It is the student's responsibility to provide the written documentation and arrange for the make-up work.

Seeking Help

The course Canvas site should be your first reference for questions about the class. If you have questions about course requirements or materials, post them on the course Canvas Discussion Board. For individual assistance, attend virtual office hours by appointment.

Late Assignments

Any assignment submitted after its deadline is considered late. Late assignments lose 10% for every 24 hours they are late. Submissions over 72 hours late are not accepted. This time limit includes weekends -- they are counted just like weekdays. I reserve the right to specify that late submissions will not be accepted for assignments.

- 0-24 hours late: -10%
- 25-48 hours late: -20%
- 49-72 hours late: -30%
- over 72 hours late: not accepted

Course Disclaimer

The course schedule and activities are subject to change. Changes will be posted as Announcements in Canvas.

AI Usage

The use of ChatGPT or other AI tools is permitted to some extent in this class. These tools are being rapidly adopted, so it is important that you have some experience with their use. These tools are best used to help you work smarter, not do your work for you. Remember that an essential part of being in graduate school is to develop skills that you will need to be successful in the workplace. Using aids just to complete an assignment in the quest for high marks will not help you after you leave school. I encourage you to use these tools to help deepen your understanding and to review material you find challenging. If you use them to blindly do your homework for you, your learning will suffer and it will be obvious. However, using them to clarify concepts, explain errors, or review material can help you learn more effectively.

Students may not submit an entire assignment, homework question, or programming task to a generative AI tool and ask it to produce the answer or solution. AI tools should be used only as learning aids (e.g., asking for explanations of concepts, debugging hints, or clarifications), not as a replacement for your own work. For all homework assignments, you must include a list of websites or other references that you consult in solving the assignment. This includes AI tools. You must include the website for the tool and either a link to, or screenshot of, the conversation you had with the tool. You must also write out in your homework report the initial prompt that you used.

*Note that this policy applies **only to CS 432/532** and does not imply that generative AI tools are permitted in other courses*

University Policies

Code of Student Conduct and Academic Integrity

The [Office of Student Conduct & Academic Integrity](#) (OSCAI) oversees the administration of the student conduct system, as outlined in the Code of Student Conduct. Old Dominion University is committed to fostering an environment that is: safe and secure, inclusive, and conducive to academic integrity, student engagement, and student success. The University expects students and student organizations/groups to uphold and abide by standards included in the Code of Student Conduct. These standards are embodied within a set of core values that include personal and academic integrity, fairness, respect, community, and responsibility.

Honor Pledge

By attending Old Dominion University, you have accepted the responsibility to abide by the Honor Pledge:

I pledge to support the honor system of Old Dominion University. I will refrain from any form of academic dishonesty or deception, such as cheating or plagiarism. I am aware that as a member of the academic community it is my responsibility to turn in all suspected violations of the Honor Code. I will report to a hearing if summoned.

Discrimination Policy

The purpose of this policy is to establish uniform guidelines in order to promote a work and education environment that is free from harassment and discrimination, as defined below, and to affirm the University's commitment to foster an environment that emphasizes the dignity and worth of every member of the Old Dominion University community. The [Discrimination Policy](#) details the process to address complaints or reports of retaliation, as defined by this policy.

Diversity and Inclusion

The [Division of Student Engagement & Enrollment Services](#) values the uniqueness of our Monarch community. The word "engagement" reflects our commitment to embrace the differences in our cultural backgrounds, perceptions, beliefs, traditions, world views, socio-economic status, cognitive and physical abilities.

We will strive to serve as the pre-eminent model for engaging every student to achieve their own success. Our core values are fueled by our responsibility and actions toward community development and engagement, cultural competence and understanding, physical and mental wellness and inclusion for every member of ODU. We will embrace our greatest strength - the diverse composition of our student body and workforce. For more information regarding diversity and inclusion, please visit the [Office of Intercultural Relations](#).

Educational Accessibility and Accommodations

Old Dominion University is committed to ensuring equal access to all qualified students with disabilities in accordance with the Americans with Disabilities Act. The [Office of Educational Accessibility](#) (OEA) is the campus office that works with students who have disabilities to provide and/or arrange reasonable accommodations.

The [Accommodations for Students with Disabilities](#) define the procedures used to accommodate student with disabilities. Students are encouraged to self-disclose disabilities that the Office of Educational Accessibility has verified by providing Accommodation Letters to their instructors early in the semester to start receiving

accommodations. Accommodations will not be made until the Accommodation Letters are provided to instructors each semester.

University Email Policy

With the increasing reliance and acceptance of electronic communication, email is considered an official means for University communication. Old Dominion University provides each student an email account for the purposes of teaching and learning, research, administration, and service. It is the responsibility of every eligible student to activate MIDAS, the Monarch Identification and Authorization System, to obtain email access. It is important that all students are aware of the expectations associated with email use as outlined in the [Student Email Standard](#). The email account provided by the University is considered to be an official point of contact for correspondence. Students are expected to check their official e-mail account on a frequent and consistent basis to stay current with University communications. Mail sent to the ODU email address may include notification of University-related actions, including academic, financial, and disciplinary actions. For more information about student email, please visit [Student Computing](#).

Withdrawal

A syllabus constitutes an agreement between the student and the course instructor about course requirements. Participation in this course indicates your acceptance of its teaching focus, requirements, and policies. Please review the syllabus and the course requirements as soon as possible. If you believe that the nature of this course does not meet your interests, needs or expectations, if you are not prepared for the amount of work involved – or if you anticipate assignment deadlines or abiding by the course policies will constitute an unacceptable hardship for you – you should drop the course by the drop/add deadline, which is listed in the [ODU Schedule of Classes](#). For more information, please visit the [Office of the University Registrar](#).

Privacy of Student Information

Old Dominion University recognizes its duty to uphold the public's trust and confidence, not only in following laws and regulations, but in following high standards of ethical behavior. Members of the Old Dominion University community are responsible for maintaining the highest ethical standards and principles of integrity. The [Code of Ethics](#) is a set of values-based statements that demonstrate the University's commitment to this goal. The [Privacy of Student Information](#) details Family Educational Rights & Privacy Act (FERPA), along with other information regarding privacy.

Statement from ODU Counseling Services

ODU's [Office of Counseling Services](#) (OCS, 1526 Webb University Center) is a university agency with competent, diverse, and multidisciplinary professional staff. OCS is committed to supporting the emotional well-being, social development, and academic progress of all ODU students.

College can be a rewarding time of self-discovery, but it may also be a time when awareness of mental health conditions increases. OCS services are available to assist students who are experiencing mental health concerns. Students can learn more about the broad range of confidential mental health services available through the [Office of Counseling Services](#). All services are free to ODU students.

Course Schedule

Each module closes on **Sunday at 11:59 PM**. All assignments are due on Sunday unless otherwise noted in Canvas or in the schedule below.

[ODU Fall 2026 academic calendar](#)

Week/Module	Date	Topics	Assignments	Due Date
0	Before Module 1	Getting Started	Required setup activities and optional extra credit activities	Multiple due dates listed in Canvas
1	10/19/2026	Introduction to Web Science and Web Architecture		
2	10/26/2026	Introduction to Python	HW1	Due: Nov 8, 2026
3	11/02/2026	Measuring & Searching the Web	HW2	Due: Nov 15, 2026
4	11/09/2026	Web Archiving & Information Visualization	HW3	Due: Nov 22, 2026
5	11/16/2026	Visualization and Analysis of Social Networks		
6	11/23/2026	Collective Intelligence & Unsupervised Learning	HW4	Due: Nov 30, 2026
7	11/30/2026	Document Filtering (Classification)	HW5	Due: Dec 10, 2026
8	12/6/2026	LLMs, kNN and Algorithm Summary		